

WINDOWS SECURITY MONITORING REPORT

Week 02 – Blue Team Cybersecurity Project

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Platform: Windows | **Primary Tool:** Windows Event Viewer

1. PROJECT OVERVIEW

This project focuses on monitoring and analyzing Windows Security Logs using Event Viewer. Important Windows Security Event IDs were studied to understand authentication, logoff, privileged activity, and account lockout events.

2. OBJECTIVE

The main objective is to understand how Windows Security Logs can be used for basic security monitoring and SOC investigation:

- Understand Windows Event Viewer and analyze Windows Security Logs.
- Identify important Security Event IDs and monitor authentication attempts.
- Analyze user logoff, privileged activity, and account lockout events.
- Develop basic SOC Analyst log analysis and correlation skills.

3. TOOLS USED

- **Windows Event Viewer:** Security log monitoring
- **Windows Security Logs:** Security event analysis
- **PowerShell:** Windows event investigation
- **Snipping Tool & VS Code:** Screenshot collection and documentation
- **Git & GitHub:** Project portfolio management

4. WINDOWS EVENT VIEWER

Windows Event Viewer is a built-in Windows tool used to monitor and analyze events generated by the operating system. Security logs accessed through Windows Logs → Security contain information related to authentication, user activity, and privileges.

5. IMPORTANT EVENT IDS

Event ID	Description	Security Importance
4624	Successful Logon	Identifies successful system authentication
4625	Failed Logon	Helps identify failed login attempts
4634	Account Logoff	Shows user session termination
4672	Special Privileges Assigned	Helps identify privileged activity
4740	Account Locked Out	Helps investigate account lockouts
4798	Local Group Enumeration	Monitors local group queries
5379	Credential Manager Read	Tracks stored credential access

6. EVENT ID ANALYSIS

6.1 Event ID 4624 – Successful Logon

Description: A user account was successfully logged on. Important fields include Account Name, Domain, Logon Type, Workstation Name, and Source Network Address. **SOC Relevance:** Helps identify who successfully logged into a Windows system and when. *Evidence: event-4624.png*

6.2 Event ID 4625 – Failed Logon

Description: A logon attempt failed. **SOC Relevance:** Repeated failed logons may indicate brute-force attempts, password spraying, or unauthorized access. *Evidence: event-4625.png*

6.3 Event ID 4634 – Account Logoff

Description: A user account was logged off. **SOC Relevance:** Helps understand when a user session ended and can be correlated with Event ID 4624. *Evidence: event-4634.png*

6.4 Event ID 4672 – Special Privileges & 6.5 Event ID 4740 – Account Lockout

Description: Special privileges assigned to new logon (4672) and user account lockout (4740).

SOC Relevance: Useful for monitoring administrative activity and investigating brute-force patterns. *Evidence:* *event-4672.png*, *event-4740.png*

6.6 Event ID 4798 & 6.7 Event ID 5379

Description: Local group membership enumeration and Credential Manager read. **SOC Relevance:**

Tracks local reconnaissance and stored credential access. *Evidence:* *event-4798.png*, *event-5379.png*

7. PRACTICAL INVESTIGATION

Opened Event Viewer, navigated to Windows Logs → Security, filtered by Event IDs, examined event details, captured screenshots, and documented findings.

8. SOC ANALYST PERSPECTIVE & FINDINGS

Investigation Chain: 4625 (Failed Logins) → 4624 (Successful Login) → 4672 (Privileged Logon). A single event does not automatically mean an attack occurred; analysts must correlate username, time, source, logon type, and normal user behavior.

Findings & Screenshots: Successfully demonstrated monitoring of authentication, logoff, privileged activity, and lockouts. Collected all required screenshots (*event-viewer.png*, *security-log.png*, *event-4624.png* to *event-5379.png*).

9. PROJECT STRUCTURE, CONCLUSION & FUTURE SCOPE

Structure: *README.md*, *screenshots/*, *event-ids/*, and report PDF.

Conclusion & Future Scope: Provided practical experience in Windows Security Monitoring and Event Viewer log analysis. Future work will include SIEM integration (Wazuh), centralized log collection, automated detection rules, and incident response.